

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

Show ip route OSPF

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? _Yes_

Does R2 show up as an OSPF neighbor on R3? _No_

What does this information tell you?

R2 is not sending OSPF Hello messages out of its S0/0/1 interface, so it cannot form an adjacency with R3.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

 enable configure terminal router ospf 1 no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

 65

Is R2 listed as an OSPF neighbor to R3? _Yes_

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

Changing the reference bandwidth allows you to influence path selection by making higher-speed links appear more attractive (lower cost), especially in networks with mixed interface speeds.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

192.168.3.0/24: R1 S0/0/1 + R3 G0/0 ($781+1 = 782$)

192.168.23.0/30: R1 S0/0/1 + R3 S0/0/1 ($781+64 = 845$)

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

$781 \times 2 = 1562$

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

OSPF selects the path with the lowest cost. The cost of the path via R2 is R1 S0/0/0 (781) + R2 S0/0/1 (781) + R3 G0/0 (1) = 1563, while the cost of the path directly from R1 S0/0/1 to R3 is R1 S0/0/1 (781) + R3 G0/0 (1) = 1566. Therefore, the lower cost path of 1563 is selected.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

Router IDs must be unique and used to identify the router and for DR/BDR election. If a dynamic interface IP is used as the RID, the RID will change when the interface fails, causing the OSPF process to reset and affect network stability. It is recommended to use a loopback interface with a static RID.

2. Why is the DR/BDR election process not a concern in this lab?

The DR/BDR election process is only an issue on multi-access network.
The serial links in the above example are the point to point. Then no DR/BDR election.

3. Why would you want to set an OSPF interface to passive?

To stop sending Hello packets on that interface, reducing unnecessary traffic while still advertising the connected network.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

The use of **areas** (especially area 0 as the backbone) allows route summarization and limits link-state updates, improving scalability.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

`network 10.0.0.0 0.255.255.255 area 0`

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The EIGRP route, because it has the lowest administrative distance ($90 < 110 < 120$).